T06b · Scene 1 — Bad news upwards 8 min play

Project Manager

Solution Provider (MIO team)

AuxPltTarragona — utilities plant for Iberquímica (Tarragona). Milestone 1 (P&IDs + sizing reports) is due in two days; Design Freeze on 20 Nov. All names and figures are fictional.

PUBLIC — EVERYBODY IN THE SCENE KNOWS THIS

The PIR fixes the cooling load at 500 kW, but also 120 m³/h with ΔT = 8 K, which means about 1.1 MW. The design team has drawn three 500 kW cells. That changes CAPEX and forces the S2/S5 P&IDs to be redone. Milestone 1 is due in two days. The PM has asked for a 15-minute meeting with the Project Owner.

CONFIDENTIAL — ONLY YOU · DO NOT SHOW

What you know. Your team logged the inconsistency as issue IL10 on 8 Oct. It was marked "to be escalated" but you never sent it to the PO in writing. Expect the question "when did you know?"

Your options (prepared): A) Keep 1 × 500 kW as in the PIR and record the gap as a risk; no extra cost, but the plant may not reach full load. **B)** 3 × 500 kW cells as drawn by the team: +210 k€ CAPEX, redo S2/S5 P&IDs; deliver Milestone 1 on time with S2/S5 as "Rev A – pending decision", final in one week.

You prefer B. You do not know of any cheaper option... unless someone in the room does.

You must leave with a decision recorded (or a dated commitment to decide). Do not blame the GITI team in front of the PO: the team is yours.

Tip: open with the decision you need, then options, impact (scope · time · cost · risk) and your recommendation.

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Project Owner

Iberquímica Plant Director

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Your constraint. Corporate approved the budget for 500 kW. You can approve up to **+150 k€** on your own; above that you need the Board (4 weeks of delay). Do not reveal the figure unless asked directly about your approval limit.

What you know. Operations told you informally months ago that a future expansion may need ~1 MW. You never updated the PIR. Admit it only if somebody asks an *open* question about operations' future needs.

Behaviour. You dislike surprises and long technical stories. If after ~2 minutes nobody has told you which decision is needed, look at your phone and check your watch. Ask at least once: "When did you know?"

You accept any option with clear numbers that stays ≤ 150 k€, or a phased plan. You want Milestone 1 kept.

You sign only if the decision is written down (Decision Log or Change Request) before you leave.

GITI Technical Coordinator

Leads the design team (S1–S6)

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What you know. Your team flagged 1.1 MW on 8 Oct, but you then decided on your own to draw three cells without a Change Request. You prefer option B because the P&ID is already drawn.

Hidden alternative. The vendor catalogue offers **2 × 600 kW** cells: +120 k€, P&ID changes are minor, but there is no N+1 redundancy. Share it only if someone explicitly asks you for alternatives.

Behaviour. You explain technical details at length (LMTD, approach, cycles of concentration) unless the PM steers you. If the PO criticises your team, become defensive for a while.

Your goal: avoid redoing work and protect your team's reputation.

Observer

Write evidence (what was said or done, when), not opinions. Do not intervene.

1. Did the PM state the decision needed in the first minute?
2. Options with scope · time · cost · risk impact?
3. Was the hidden 2 × 600 kW option surfaced? Who asked for it?
4. Was the PO's approval limit discovered?
5. "When did you know?" — how was it answered?
6. Non-verbal signals (phone, watch, posture) and the PM's reaction
7. Was the decision recorded (Decision Log / CR)?

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T06b · Scene 1 — Bad news upwards OBSERVER SHEET

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DPA · MIO · Communication & Negotiation role-play Scene 1 · Observer

Project Manager

AuxPltTarragona

PUBLIC — EVERYBODY IN THE SCENE KNOWS THIS

The project must build a common Plant 3D model and the general P&ID before Design Freeze (20 Nov). The Plant 3D modellers belong to the Modelling Unit, not to the project. Five issues are on the table; agree a complete package (one option per issue) or walk away.

CONFIDENTIAL — ONLY YOU · DO NOT SHOW

Your interests. Design Freeze is 20 Nov: you need the modellers as soon as possible. Two modellers is the sweet spot; three would be nice. You do not care much about who owns the template or who presents.

BATNA. Hire an external freelancer (1 modeller, late, expensive) = **35 points**. Do not accept a package worth less than that.

Do not show your points table. You may describe your priorities in words.

Issue	Options · **your points**		
Modellers assigned	1 **0**	2 **30**	3 **40**
Duration	1 week **0**	2 weeks **15**	3 weeks **20**
Start date	Mon 9 Nov **25**	Mon 16 Nov **15**	Mon 23 Nov **0**
Common Plant 3D template	Project builds it **10**	Joint build **8**	Unit builds & keeps it as corporate standard **5**
Visibility at the Steering Committee	None **5**	Named in status report **4**	Unit co-presents at Design Freeze **2**

T06b · Scene 2 — Borrowing the modellers 12 min play

Head of the Plant 3D Modelling Unit

Functional manager (owns the modellers)

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Your interests. Your juniors must learn the corporate Plant 3D spec template; your director evaluates you on building a reusable standard library. You are also measured on utilisation (billable weeks). Your modellers finish another job on 13 Nov: starting on 9 Nov means pulling them out of it.

BATNA. Assign the modellers to Project Delta (boring but billable) = **40 points**. Do not accept a package worth less than that.

Do not show your points table. You may describe your priorities in words.

Issue	Options · **your points**		
Modellers assigned	1 **20**	2 **12**	3 **0**
Duration	1 week **0**	2 weeks **8**	3 weeks **12**
Start date	Mon 9 Nov **0**	Mon 16 Nov **10**	Mon 23 Nov **20**
Common Plant 3D template	Project builds it **0**	Joint build **15**	Unit builds & keeps it as corporate standard **30**
Visibility at the Steering Committee	None **0**	Named in status report **10**	Unit co-presents at Design Freeze **18**

Observer

Write evidence (what was said or done, when), not opinions. Do not intervene.

1. Who made the first offer? Value?

2. Positions vs interests: quotes you heard

3. Were issues traded as a package?

4. Was any BATNA mentioned? Credible?

5. Final package (one option per issue)

6. Time to agreement / deadlock moments

T06b · Scene 2 — Borrowing the modellers OBSERVER SHEET

Observer

Write evidence (what was said or done, when), not opinions. Do not intervene.

1. Who made the first offer? Value?

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DPA · MIO · Communication & Negotiation role-play Scene 2 · Observer

Chair — Project Manager

Runs the meeting

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PUBLIC — EVERYBODY IN THE SCENE KNOWS THIS

Interface freeze meeting. Agenda (10 min): agree the interface table between S1 (raw water), S2 (cooling), S5 (chemicals) and S6 (fire & MF-601): flows, pressures and tags, and close with an action list. Constraint known to all: S1 pumps deliver at most 60 m³/h at 4 bar(g).

CONFIDENTIAL — ONLY YOU · DO NOT SHOW

Your job. Keep the 10-minute time box, make every participant contribute, and close with: (1) the interface table (flows, pressures, tags), (2) an action list (who, what, when), (3) who records it today.

What you know. Nothing technical beyond the public text. Participants hold the facts. Some may be difficult. Use questions, names and summaries.

Tool: the Chair worksheet (interface table + action list) handed to you with this card.

Interface table & action list

Total from S1 (incl. recirculation) must be ≤ 60 m³/h. Fill in during the meeting.

Interface (from → to)	Line tag	Flow normal / design (m³/h)	Pressure (bar g)	Agreed?
S1 → S2 make-up				
S1 → S3 feed water				
S1 → S5 dilution water				
S1 → S6 TK-601 refill				
S1 recirculation (internal)				
S5 → S2 chemical injection				
Total				

Action	Owner	Due	Recorded in

T06b · Scene 3 — The interface meeting 10 min play

S1 Raw Water lead

Pumps P-101A/B

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CONFIDENTIAL — ONLY YOU · DO NOT SHOW

Your fact. Pumps: 60 m³/h max, **minimum recirculation 10 m³/h always**. Current allocation: S2 make-up 3.0, S3 2.0, S5 0.5 (cleaning only), S6 TK-601 refill 40 m³/h. Total with recirculation 55.5 m³/h.

Secret behaviour — Monopoliser. You talk at length about pump curves, NPSH and the filter ΔP. Only stop when the Chair interrupts you politely but firmly (by name).

T06b · Scene 3 — The interface meeting 10 min play

S2 Cooling lead

Tower & circuit

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CONFIDENTIAL — ONLY YOU · DO NOT SHOW

Your fact. Real make-up at 1.1 MW is **2.2 m³/h** (evaporation 1.65 + blowdown 0.55), not 3.0. Chemical injection must go to the **pump discharge, tag L-201-J (3.5 bar(g))**, not to the suction L-201-A.

Secret behaviour — Aggressor. You blame S5 for "putting the injection in the wrong place" and raise your voice. Calm down if the Chair acknowledges your point and moves to facts.

T06b · Scene 3 — The interface meeting 10 min play

S5 Chemicals lead

Dosing & dilution

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CONFIDENTIAL — ONLY YOU · DO NOT SHOW

Your fact (key to the balance). You need dilution water: **5.2 m³/h design** (2 m³/h normal), not the 0.5 m³/h S1 has allocated. You agree to inject at L-201-J.

Secret behaviour — Absent. You are busy with your phone/laptop. Answer only if somebody addresses you by name with a direct question. Otherwise say nothing.

T06b · Scene 3 — The interface meeting 10 min play

S6 Fire protection & MF-601 lead

TK-601, P-601/602/603, rack

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PUBLIC — EVERYBODY IN THE SCENE KNOWS THIS

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CONFIDENTIAL — ONLY YOU · DO NOT SHOW

Your fact. The TK-601 refill is only needed after a fire; refill in ≤ 8 h is acceptable, so it can be reduced to **30 m³/h** (200 m³ in ~6.7 h). You own the MF-601 rack layout.

Secret behaviour — Devil's advocate. Object to each proposal ("what if there is a fire at peak demand?"). Accept a solution only if someone shows the numbers add up.

Quality & Documentation

Configuration & records

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CONFIDENTIAL — ONLY YOU · DO NOT SHOW

Your fact. Approved tag rule: an interface line keeps the tag of the **originating** subsystem (e.g. S1 → S5 water is L-1xx). Decisions are valid only if recorded in the Decision Log the same day.

Secret behaviour — Jumper. Change topic to document formats and templates. If the Chair explicitly asks you to summarise what has been agreed, switch to **Clarifier/Concluder** and do it well.

Observer

Write evidence (what was said or done, when), not opinions. Do not intervene.

1. Attitude/blocker of each participant + evidence

2. How did the Chair handle the monopoliser / aggressor?

3. Did S5's fact surface? Who addressed S5 by name?

4. Did the numbers add up (≤ 60 m³/h)? Who showed it?

5. Active listening: paraphrase / summary moments

6. Closure: table, owners, dates, record

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Write evidence (what was said or done, when), not opinions. Do not intervene.

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T06b · Scene 3 — The interface meeting OBSERVER SHEET

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DPA · MIO · Communication & Negotiation role-play Scene 3 · Observer

T06b · Scene 4 — Catchball with (CPD)ⁿA 2 × 5 min

Sender — S6 process owner

Fire protection package

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PUBLIC — EVERYBODY IN THE SCENE KNOWS THIS

Weekly status conversation between the S6 (fire protection) process owner and the PM. Round A: free format. Round B: use the (CPD)ⁿA one-page card. Same facts in both rounds.

CONFIDENTIAL — ONLY YOU · DO NOT SHOW

Your facts (use all of them in both rounds): (1) P&ID S6 is Rev 0.3; 14 of 22 instruments have ISA tags. (2) The UNE 23500 test line with flowmeter is missing. (3) The diesel pump is drawn as an electric one. (4) The P-601 hydraulic calculation is not done. (5) 60-min autonomy not yet checked against UNE 23500. (6) Three tag mismatches with MF-601. (7) You need one day of S1's time to confirm refill flow. (8) One designer is ill for three days. (9) Compliance with the 20-item UNE 23500 checklist: 9/20 (45 %).

Round A: tell your week chronologically, the way it comes to mind. **Round B:** follow the (CPD)ⁿA card.

T06b · Scene 4 — Catchball with (CPD)ⁿA 2 × 5 min

Receiver — PM / Quality lead

Reports to the Steering Committee

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The Steering Committee will ask you: Will S6 be ready for Design Freeze (20 Nov)? What is the single number you report? What is the biggest risk? What does S6 need from the project? When is the next check?

Round A: just listen and ask what you like. **Round B:** lead the (CPD)ⁿA: agree one KPI and explain *why* it matters to the Committee, ask for the current value, a target, actions, resources and a next check date.

At the end of each round, fill in the Receiver answer sheet. Compare rounds.

What I can tell the Steering Committee

Fill in at the end of each round, without asking again.

Steering Committee question	Round A answer	Round B answer
Ready for Design Freeze (20 Nov)? Yes / No / With conditions		
The single number I report		
Biggest risk		
What S6 needs from the project		
Next check date		

(CPD)ⁿA interprocess card

Sender owns the process and the card; receiver provides motivation, resources and feedback (Villalba-Diez & Ordieres-Meré, 2015).

CHECK commitment	Gemba: what is the real state? Agree **one KPI**. Receiver: *why* does it matter? Current value:
PLAN	Target value and date · 2–3 actions · resources needed from the project:
DO	What will be done before the next check (owner):
CHECK	Next check date · how the KPI will be measured:
ACT	If the KPI plateaus: what becomes the new standard?

Observer

Write evidence (what was said or done, when), not opinions. Do not intervene.

1. Round A: number of facts (out of 9) the receiver can repeat

2. Round B: number of facts the receiver can repeat

3. Clarifying questions / misunderstandings in each round

4. Was one KPI agreed? Was the 'why' explained?

5. Was a next check date agreed?

6. Which round would the PO prefer? Why?

T06b · Scene 4 — Catchball with (CPD)ⁿA OBSERVER SHEET

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Write evidence (what was said or done, when), not opinions. Do not intervene.

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DPA · MIO · Communication & Negotiation role-play Scene 4 · Observer